5. Design of the Workplace

General Work Station Design Principles

After the industrial hygiene engineer has designed the job to meet the criteria of the desired output and provided the appropriate tools and procedures to protect the worker from potential hazards, it is necessary to design the work station in which the worker will perform the job. The general principle that should be followed on work station design is to provide an efficient and safe location in which the work can be performed. There are certain basic factors that must be considered when designing a workplace.

First, the designer must determine what the worker must see while on the job. What part of the work environment must be visible to the worker? While performing the job, what controls must the worker operate? Where should these controls be located, considering the type of operation that is necessary? Is it necessary for the worker to be able to view the action of the equipment being used on the job? What about the interaction of the job with other workers and their jobs? Is it necessary for the worker to be able to view other jobs and worker actions?

Third, the designer should consider what operations the worker must perform while on the job. Is it necessary for the worker to operate hand and/or foot controls? Will the worker be required to do lifting and carrying? While on the job, does the worker need to position materials in order to accomplish the task? What motions are required while performing the job? What special tools are used to perform the job? Are physical demands other than material handling placed upon the worker by the job?

The designer should consider the job procedure itself. What is the sequence of operation performed on the job? Is there any priority of operations that the worker must perform?

Important in the design of the workplace is the determination of clearances that are required to perform the job. Such clearances should consider the size of the workers who will perform in the workplace and include clearances for movement. Is there any chance for accidental injury by bumping objects in the workplace? The designer should consider the possibility of accidental activation of controls. What is the likelihood of such an event occurring and how can it be prevented?

Finally, the designer should consider what storage requirements are necessary for the job. Provision must be made for the storage of raw materials, in-process materials, and finished product. Also, the designer

must consider what hand tools or other aids the worker will use that must be stored in the workplace.

Design for Visibility and Hearing

When designing the workplace for visibility and hearing, the first problem that the designer must face is where to locate the displays and controls that must be monitored and used by the operator. Displays and controls should be located in such a manner that the operator can perform the necessary monitoring and adjustment with a minimum disruption to other duties being performed on the job.

The first priority is the visual requirement of the job, and this requirement determines the general layout of the workplace. The operator should be positioned in order to face in the direction of the primary visual field. If it is necessary for the operator to view the operation of equipment, the operator should be faced in that direction with a clear and unobstructed field of vision between the operator and the equipment being viewed. If, as a part of the job, the operator must view certain displays or operate certain controls continuously, then these should be placed within the primary field of view.

The second priority is to locate the primary controls acting on the primary visual task. If, for example, the primary visual task is operating a crane to lift objects and it is necessary for the operator to view the crane operation while controlling the crane, the primary controls for operating the crane should be located in such a manner that the operator does not have to move positions from the primary viewing task.

Any emergency controls that are visual in nature should be located close to the primary field of vision. The operator should be able to observe any warning indications, such as blinking lights or pointers indicating danger, while viewing the primary task.

Displays that are related to the primary controls should be within or as close to the primary field of vision as possible. Where it is not possible to locate such displays in the direct field of vision, they should be located in such a manner that will require only the movement of the operator's eyes to observe them. Again using the automobile for an example, the speedometer and rear view mirror, though not located in the primary field of vision, are located where a quick glance of the eyes can ascertain the desired information.

Secondary controls or displays that are referred to infrequently can be placed in an out-of-the-way area. However, as much as possible, such controls should not require the operator to change body positions since this may interfere with the primary viewing task.

Auditory signals should be located in reference to the action causing the signal. If, for example, a warning is related to something happening to the right of the workplace, it is preferable to place the auditory signal on the right of the operator. This will automatically direct the worker's attention to that area of the workplace. The auditory signal itself should be pointed

to the listener's ear and be close enough to overcome any noise masking that might occur in the workplace.

Finally, visual controls and displays should be grouped functionally. In this manner, it is possible for the worker to check a number of related controls and displays during one observation.

Once the controls and displays have been located, it may be necessary to label or code these controls and displays in some manner. One method of labeling is to color code the displays and controls. Generally, red is used for emergency situations, while yellow is often used for cautionary situations; and green is used for safe or normal operation.

The controls can be labeled if it is possible for the worker to observe the control panel while operating the equipment. If labels are used, they should be placed consistently relative to the controls. For example, all controls should be placed either above, below, to the right, or to the left of the control to which they refer.

It is useful to color the knobs of the controls differently than the background panel. This helps the operator to see quickly the actual location of the knob. Functional groups of controls can be offset in separate areas. and the background panel shaded or lined off to indicate the functional group. Next, the designer should consider the amount of illumination that is present in the workplace. Is the illumination provided in the workplace adequate for the tasks required? If the tasks require fine adjustments and/or assembly, the illumination should be sufficient for the operator to discriminate between the fine differences in adjustment. The designer should take care to assure that glare will not be a problem while the worker is performing the job. Shiny surfaces are always subject to producing glare in the operator's eyes, particularly if the operator finds it necessary to move his position to perform certain tasks. In such cases, matte finished panels and glare resistant glass are useful.

Finally, the designer should consider the amount of illumination entering from outside the work area. In some cases, problems related to glare and visibility can result from the entrance of sunlight or floodlights from other areas. Where windows are located near the workplace, consider the angle of the sun during the work period throughout the year to determine if at any time the entry of sunlight will cause a visual problem.

Design for Worker Operations

While the worker is in the workplace, it may be necessary to perform certain physical or manual tasks. Certain types of equipment may also be required to perform the job. During the performance of the job, different motions will be required. The designer must design the workplace in such a manner as to provide for the motions that are required to operate the equipment. An additional consideration is the placement of the equipment itself. It should be placed in such a manner that it is easily accessible to the operator.

Of major consideration in the design of the workplace is the operator's position within it. Should the operator work from a sitting position, standing position, or a combined sitting and standing position? Certain general rules have been derived to determine which position is best, given the type of job to be performed. The following is adapted from the book, Human Engineering Guide to Equipment Design, VanCott and Kinkade, 1972.

The sitting position is best for:

1. A high degree of operator stability.
2. Precise controlling.
3. Long work periods.
4. Use of both feet for controlling.
5. Large-force application or a range of movement for foot controls.
6. Precise foot movements.

The standing position is best for:

1. Mobility to reach and perform operations.
2. Where large areas must be monitored.

A combined sit-stand position is often required by the job. In such cases it is generally best to provide a high stool since this enables the operator to move from one position to another quickly. When a high stool is provided in such a workplace, an adjustable footrest should be attached to the stool to enable the operator to maintain the proper position while sitting.

During the design of the workplace, it is necessary to determine the actual placement of manual controls to assure efficient and nonfatiguing operation. Primary controls should be located within the area of comfortable reach. In general, this area is determined by the arc of the arm with the elbow held near the body. In other words, it is an arc with the radius equal to the length of the worker's forearm and hand.

Secondary controls can be located beyond comfortable reach; that is, in an area circumscribed by an arc with a radius equal to the length of the individual's extended arm. Seldom used controls can be located in an out-of-the-way position that may require the operator to change positions to make adjustments.

Cranks should be placed at elbow level. Pull-up controls should be located below seat level or at near full extension of the arm when the operator is standing. Pull-out controls should be just short of full-arm extension for either the seated or standing operator.

Provision for Equipment in the Workplace. Where electrical power tools are going to be used by the operator, provision should be made for adequate electrical outlets. Each of the outlets should be of the three-prong grounded variety.

The designer should determine the adequate space that is necessary to operate the equipment in its most efficient manner. A clear work area should

be available for the operation of the equipment. Consider each individual piece of equipment to be used as well as the particular operation performed by each piece of equipment. In order to develop a proper work space design, it may be necessary for the designer to observe a worker using the equipment to perform the required tasks.

The equipment used should be in logical order. Generally, such an order is according to the task performance sequence. The location in relation to the sequence should be either from left to right or top to bottom in the work station.

The designer should provide methods for stabilizing the work; such as jigs, vices, or built-in positioning frames. Such aids will enable the operator to free one hand to perform other tasks or to operate other controls.

The designer should locate the most frequently used equipment nearest the worker. Less frequently used equipment can be located in an out-of-the-way position.

Standard Design. Where similar jobs are being performed by workers in different workplaces, the designer should consider standardizing the workplace layout and design for these operations. This provides the ability to shift workers back and forth between job duties with a minimum amount of training concerning the workplace layout. In addition, it reduces errors that will occur after a worker has been moved to a new job.

Provide Adequate Storage. An important consideration in design of the workplace is to provide adequate storage for both materials and equipment. A cluttered workplace causes errors and accidents as well as being detrimental to the worker's morale. Positions for tools and materials should be near their point of use. This will assist the worker in maintaining an uncluttered workplace since such a location will make it much more likely that the tool or material will be replaced when that particular part of the job is completed. Just as we often find in our own homes or apartments that there is never enough storage space, the worker often finds that the designer has not provided adequate space to store materials and tools.

Figure 7.5.1

Storage requirements.



Workplace Space Considerations

Another consideration in the design of the workplace is how large it should be. In locating equipment within comfortable reach, what determines comfortable reach?

Generally, workplace space dimensions are based on anthropometric data that have been gathered. Normally, the workplace is designed to meet the requirements of those individuals between the fifth and ninety-fifth percentile range of the particular dimensions that are important to the job. Unless there is some known reason to vary, this approach should be used in the design of the workplace. A final checkout of data on a sample of workers may be helpful to determine the appropriateness of the anthropometric data that have been chosen.

Clearance requirements should be based on the largest dimensional data for the particular movements involved. For example, the overhead clearance should accommodate the largest person who might work in the work area. On the other hand, reach movements should be based on the smallest individual who will work in the workplace. In this manner, a maximum number of individuals will be able to perform the job efficiently.

The designer should allow for adjustability where possible. Most commonly, such adjustability can be obtained in the seating and height of the workbench. This will allow for the accommodation of both large and small workers.

When determining the space considerations, the designer should consider any clothing that will be worn by the worker that may increase the size dimensions or limit the motion of the worker. Also of importance are postural changes that may occur while the worker performs his job. As much as possible, the workplace should accommodate changes in posture, since these are necessary in order for the worker to prevent excessive fatigue.

In some cases, the work space may be in a vehicle or other moving systems; such as a crane, elevator, etc. In these situations, it is important for the designer to consider the clearance requirements in the case of a sudden stop or start. Swaying and turning are often problems in moving vehicles and should be considered for clearance requirements by the designer.

The workplace should be designed to provide a safe and healthful environment in which the worker can perform tasks. In order to do this, corners and angles that may be bumped by the worker should be round. Any projections should be removed as much as possible. Assurance should be made that the controls are protected from accidental bumping during the performance of other tasks, thus inadvertently activating equipment that could cause an accident or injury to the worker. Where the workplace is in motion, restraints--such as seat belts, guardrails, and other such devices--should be provided to prevent the worker from being moved from the normal work position.

A poorly designed work space can cause physical stress on the worker's body. Fatigue can set in if the worker is required to work in an awkward posture. The seat and workbench should be designed in such a manner as to encourage proper work posture. Certain types of seating can cause a dangerous limitation of blood flow to the lower extremities of the body. This can adversely affect the health of the worker after long periods of exposure.

In addition, where the work space itself is moving or where mechanical equipment is operating in close proximity to the workplace, there is the potential for vibration being transmitted to the worker. This vibration can be harmful to the worker if the exposure continues for an extended period of time. Methods should be provided to limit this vibration as much as possible.

The Design of the Plant Equipment Layout

The objective of plant layout is to locate the individual work stations within the plant such as to provide for efficient production, provide for efficient material flow, and provide a safe environment. Plant layout becomes important either when a new plant is being designed or when a renovation program of an existing plant is being considered. The industrial hygiene engineer should take an active part as a member of the task force charged with the plant layout. The involvement of the industrial hygiene engineer should be early in the design stage since the factors that are important for employee health and safety are most often related to major groupings of equipment. If the industrial hygiene engineer is involved in the design process, potential hazards to the health and safety of the workers can be prevented.

Generally the process layout is used to determine the general plant layout design. With such a layout, machines are arranged in functional groups. Material flow throughout the plant is then charted. The objective of such a layout is to reduce material handling costs and provide for efficient operation. Generally a layout that is efficient from a material handling viewpoint will provide for the safety of the worker since many of the accidents and injuries that occur on the job are as a result of material handling activities.

Where material handling activities cross paths going between functional areas of the plant, hazards exist. Also, inefficient material handling can make it difficult to meet production quotas, thus requiring that the workers within the plant operate at higher than safe speeds. Imagine a plant where fork trucks carrying materials are zipping to and fro randomly between operations. Such a plant layout is certainly highly likely to be unsafe.

In addition to material handling, other factors must be considered in the design and layout of the plant. In general, hot processes should be grouped together and isolated to limit the exposure to only a few workers. Noisy operations should be planned carefully and where possible isolated from the remainder of the plant. Noncompatible operations where there is a potential for hazardous materials to mingle should also be located in separate areas of the plant. In no case should operations requiring local exhaust ventilation be mixed in the same area with operations requiring general or dilution ventilation.

When looking at the location of each individual workplace within the plant, the designer should consider the protection of the worker and the equipment. Guardrails should be provided to eliminate the possibility of the operator entering the traffic area or vice versa. Adequate areas for

maintenance of equipment should also be provided. Remember, it is the duty of the industrial hygiene engineer to protect the maintenance worker as well as the production worker. Space should be allowed around the work area for storage of raw materials, work in process, and finished materials; and these should be in such a location so as not to interfere with the normal operation of the equipment.

Use of Color Coding in the Plant. One method that is available to provide information concerning the potential hazards within the plant is to color code areas, equipment, and piping. Standard color codes are generally accepted for such coding. To code areas of the plant and equipment, the following codes are generally accepted:

1. Red--Indicates a danger or fire protection equipment.
2. Yellow--Indicates an area where caution should be exercised.
3. Green--Indicates the location of first aid equipment.
4. Black and White Stripes or Checks--Generally used for traffic markings, particularly where mobile traffic is present.
5. Orange--Generally used to indicate hazardous parts of machinery where the operator must exert extreme care.
6. Magenta--Used to indicate radiation areas.

In the same manner, piping may be color coded to indicate the contents of the pipe. This is particularly important since it is not always obvious which pipes contain which materials. The generally accepted color codes are as follows:

1. Red--Indicates fire protection materials, most often water.
2. Orange or Yellow--Indicates hazardous materials are contained within the pipe.
3. Green--Indicates safe materials are transported in the pipe.
4. Blue--Indicates protective materials are being transmitted.

Traffic Spaces

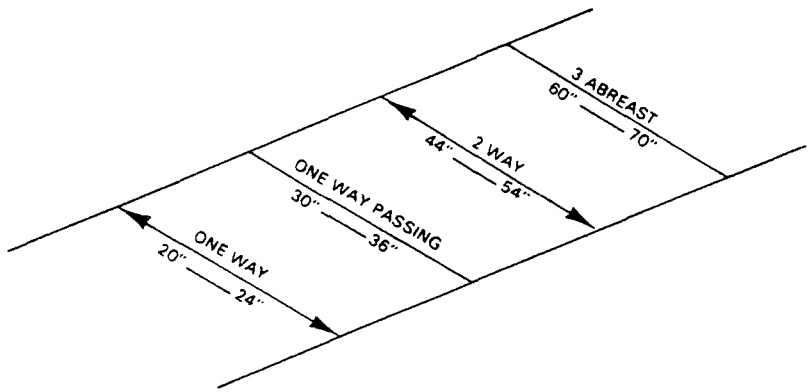
In addition to providing for appropriate location of equipment and work spaces within the plant, it is necessary to provide for the movement of traffic throughout the plant. Traffic movement involves the allowance for aisles and corridors, exits and entrances, as well as ladders, stairs, ramps, and elevators.

Aisles and Corridors. The designer should consider the purpose for which the aisle or corridor is to be used. Will one-way traffic likely exist in the area, or is it necessary to provide two-way traffic? Will the traffic be essentially pedestrian traffic, or will it be mechanical equipment; such as fork trucks and tractors pulling carts? In general, the dimensions shown in Figure 7.5.2 are recommended for aisles and corridors that will handle pedestrian traffic. The data presented in the figure are based on anthropometric data.

The designer should take into consideration other factors when designing aisles and corridors. As much as possible, blind corners should be avoided

Figure 7.5.2

Aisles and corridors.



since there is the potential for accident or injury. This is particularly the case when mechanical vehicles are using the aisles. The use of stop signs and mirrors to indicate oncoming traffic around the corner can be helpful.

The designer should determine where dense traffic is likely to occur, both in the normal working day and in times of emergency. The design should provide for the maximum traffic expected in the area.

Aisles should be marked clearly to indicate the area in which traffic should be constrained. The aisles should be kept clear of any obstructions. This includes doors, which should not open into the corridor.

Corridors should be intersected at ninety degrees where possible. This will conserve valuable space that can be used in the actual production processes.

Exits and Entrances. It is important to provide adequate exits and entrances to the facility or the various areas of the facility. Emergency exits should be located to provide exit from the work area in case of emergency. Multiple exit points are desirable to eliminate crowding at any one exit and to provide alternate choices in case the emergency eliminates one or more of the exits. A check should be made of the local building codes so that sufficient exits are provided and properly located in order to be in compliance.

In the case of emergency exits, the door should open outward. A panic bar should be installed to assure that the push of the crowd will open the door easily. In the case of an emergency, a jammed door can cause serious injury and perhaps death to those trapped inside.

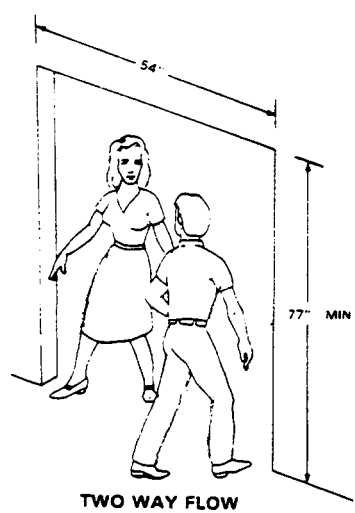
In general, ninety-degree corridors should be avoided for emergency exits. The straight line emergency exit is best. Where an intersection is necessary, the intersection should be from one aisle into two rather than from two aisles into one. This eliminates the possibility of congestion occurring, resulting in possible injury to the workers.

When designing the doors to be used as exits, the revolving door or spring-loaded swinging door should be avoided. These doors can cause unnecessary accidents or injury. If double doors are to be installed, each door should operate in only one direction. Where glass doors or panels are used, the glass should be frosted or marked with a pattern in order to prevent individuals from inadvertently assuming that the door is open and walking into the glass.

Figure 7.5.3 presents the dimensions for human traffic through entrances and exits for two-way flow.

Figure 7.5.3

Entrance and exit.



Consideration should be given to the protection of individuals standing or walking near a hinged door. One method for accomplishing this is to place a glass window halfway up so that individuals on both sides of the door can view others approaching the door. Where possible, the door should open into a dead space or towards a wall. In any case, space should be allowed between the arc of the door and any equipment in order that a person can move out of the way of the swinging door and avoid injury.

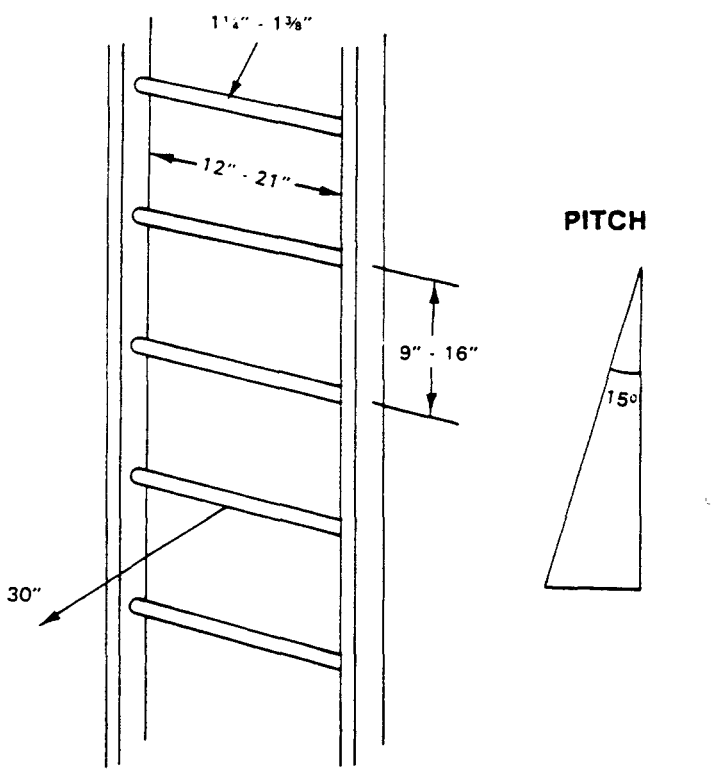
Ladders, Stairs, and Ramps

The pitch of fixed ladders should be between 60 and 90 degrees. It is preferred that the pitch be in the area of 70 to 90 degrees. The rung diameter of the ladder should be between 1-1/4 and 1-3/8 inches with a distance between rungs of between 9 and 16 inches. The distance between side rails of a ladder should be between 12 and 21 inches, and the distance to the nearest object in front of the ladder should be 30 inches if the ladder is at a 90° angle and 36 inches if the ladder is at a 75° angle. Where ladders are located between floors, the ladder should be offset to prevent the possibility of the individual falling more than one floor.

Figure 7.5.4 shows the recommended dimensions for ladders.

Figure 7.5.4

Ladders.

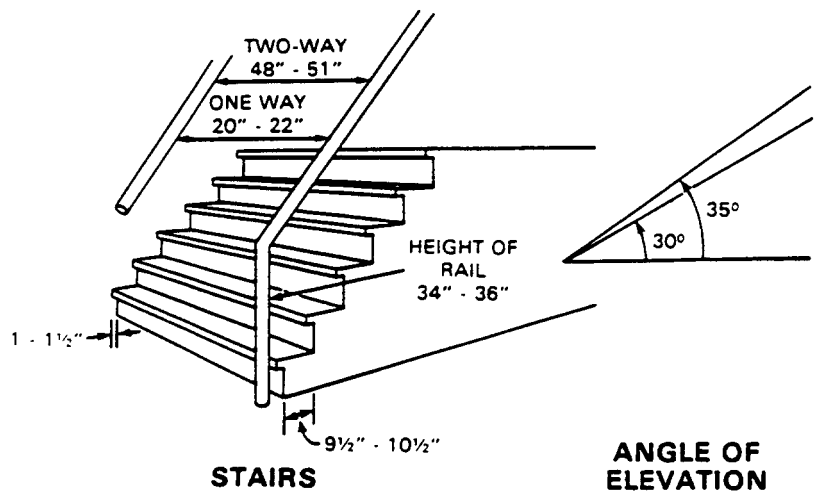


In the design of stairs, the rise between floors should be at a 20 to 35 degree angle. The riser height should be between 5 and 8 inches with the tread depth between 9-1/2 and 10-1/2 inches.

Where possible, landings should be provided every 10 to 12 steps. Stairs should be approximately 20 to 22 inches wide to allow for one-way traffic and 48 inches to 51 inches wide to allow for two-way traffic.

Figure 7.5.5

Stairways.



Handrails should be provided. These handrails should be at a height of 36 inches above the stair tread. Double rails and guard screens should be provided where 90 degree landings are present. This will prevent individuals from slipping and falling off the landings.

Where a ramp is used, the maximum slope should be 20 degrees. If the slope is to be greater than 15 degrees, cleats should be used to provide protection against slipping. Handrails should be provided along the side of the ramps to allow for pedestrian use. These rails also guard against the possibility of vehicles dropping off the side of the ramp.

Summary

It is important that the industrial hygiene engineer consider the design of the workplace. The workplace should be designed in such a manner that it provides a safe and healthful environment for the worker. Certain general principles concerning the workplace and station design have been presented that should assist in accomplishing this objective. In general, design of the workplace should be based on what the worker must see, what the worker must hear, what the worker must do, and what space requirements the worker must have to accomplish the required tasks. In addition, the overall plant layout is important to provide for a healthful work environment. The layout of

equipment within the plant as well as traffic spaces must be considered. The industrial hygiene engineer must work closely as a member of the team that is charged with designing a new plant facility or the renovation of an existing facility. This involvement must begin early in the design process if it is to be effective in providing for the required safe and healthful environment.